

Novelty index of the thesis

Results that are new, that extend previous work,
and that are already in the literature

Adam Aldridge

21 August 2026

Fix_Notes (Novelty). In all the results, both presented and derived, you must emphasize those which are new, how they differ from or extend previous work, and a discussion of why they are important.

Contents

1	How to read this index	3
2	Whole-thesis verdict	4
2.1	Original statements (not found as written)	4
2.2	Real extensions (ancestor plus increment)	6
2.3	What not to claim as new	7
2.4	Counts (reconciled)	8
2.5	Citations the thesis is missing or under-using	8
2.6	Suggested sentence shapes	9
3	Chapter 1 — Introduction	10
4	Chapter 2 — Mathematical introduction	12
5	Chapter 3 — Galton–Watson and $A(p)$	15
6	Chapter 4 — BDC: definition and main results	17
7	Chapter 5 — BDC distribution, QSD, burst statistics	20
8	Chapter 6 — Burst-aware renewal and population comparison	24
9	Chapter 7 — Two-type BDC	27
10	Chapter 8 — Bursting and budding pathogenesis	30
11	Chapter 9 — Variability in evolution	33
12	Chapter 10 — Conclusion: evolved strategy of <i>Yersinia pestis</i>	35

Chapter 1

How to read this index

This document is a literature map of every extracted result in the thesis (320 items across ten chapters). It is not a rewrite of the chapters. Use it to decide what the thesis should cite as known, what it should claim as original, and why that claim matters.

Ten extractors listed every theorem, closed form, model, comparison, and argument that functions as a result. Literature agents searched the mathematical object (not the slogan) against the chapter bibliographies and the wider literature. Adversarial checkers then tried to downgrade marks of novelty. One checker opened Karlin and Tavaré (1982) in full.

Classes.

Original This formula, identity, or named construction was *not found* in the literature. Neighbours exist; the method or the question may be published. This is the strongest claim made here.

Extends A published result is the ancestor; the thesis changes the process, the object, the hypotheses, or the biological reading.

Already Equivalent statement already published, even if re-derived here.

Not a result

Recap, definition, figure, or framing.

“Not found” is not a uniqueness proof. When evidence was thin, the class was moved toward ALREADY or EXTENDS, not toward ORIGINAL. After adversarial checking, **no result is without published ancestors**. Several statements were still **not found as written**. Those, together with the ancestor and the increment, are what the Fix_Notes item wants emphasized.

Chapter 2

Whole-thesis verdict

The analytical spine of the thesis is the linear birth–death process with per-capita total catastrophe (killing into a cemetery H). That process, its Riccati / linear-fractional generating function, the geometric occupancy, the geometric killing position, and the identity “burst size equals the quasi-stationary law” are **Karlin and Tavaré (1982)**. The constant $A(p)$ of Chapter 3 is **Kolmogorov (1938)**. Constant-rate BMVR, R_0 at matched lifetime yield, and stochastic bursting-versus-budding at establishment are **Perelson / Nowak / McLean and Pearson, Krapivsky and Perelson (2011)**. Antibody flooding is **Komarova (2007)**. Diversity-dependent and time-dependent speciation alternatives to Nee’s push of the past are a large existing literature.

What the thesis actually adds is a **closed-form intracellular calculus** on top of Karlin–Tavaré, a **renewal embedding** of those kernels into the within-host model, an **L -organised comparison** of bursting and budding, a **two-type killing deformation** of Antal–Krapivsky, two **finite-budget immune caricatures**, and a cluster of **evolutionary / plague-strategy** models that are applications or unfinished specifications. Those are the results to emphasize. The 1982 / 1938 / 2011 / 2007 facts must be cited as such, or the novelty claims will not survive viva.

The four ORIGINAL statements in the strict sense (no published equivalent formula) are CH6-R20, CH6-R21, CH7-R06, and CH7-R23. Adversarial checkers would label even those EXTENDS, because Pearson+Karlin–Tavaré and Antal–Krapivsky supply the *method*. In the thesis, write them as original identities obtained by specialising those methods, not as a new theory of branching.

2.1 Original statements (not found as written)

These are the closest the map comes to “does not currently show up in the literature.” Each has a named neighbour. Write them as original *statements*, with the neighbour cited in the same sentence.

ID	Short name	Class	Nearest literature
CH6-R20	Flooding identity $\Delta z_{\text{ext}} = (L - 1)(1/m - 1)$ and criterion $L > 1 \Leftrightarrow 1/\delta < 1/\lambda + 1/\mu$	ORIGINAL	Pearson 2011 (arbitrary burst law vs geometric drip; no L); Quirouette 2023 (law, not timing)
CH6-R21	At $L = 1$ the bursting and budding offspring <i>laws</i> coincide (not just extinction probabilities); variance order flips with $L - 1$	ORIGINAL	Pearson geometric drip; Karlin–Tavaré geometric kill-size; Quirouette

continued

ID	Short name	Class	Nearest literature
CH7-R06	Exact finite-time no-catastrophe $S(t)$ for triangular two-type BD with type-specific linear catastrophe $\delta_1 X + \delta_2 Y$, Wronskian mix of a real-axis ${}_2F_1$ pair	ORIGINAL	Antal–Krapivsky 2011 (same Gauss reduction, terminal-count PGF, no catastrophe). $\delta = 0$ in the survival IVP does <i>not</i> recover AK
CH7-R23	Killed (defective) bivariate PGF by the same mix; AK is this object at $\delta_i = 0$	ORIGINAL	AK 2011 (25)–(29)
CH4-R17–R20, R22	Finite-time dump W_t : $V(t)$, $\mathbb{E}[W_t^2]$, $\text{Var}(W_t)$, current yield $\mathbb{E}[X_t + W_t]$ as functions of $I(t)$	EXTENDS	KT 1982 killing position is the <i>eventual</i> geometric; not a time-dependent dump coordinate
CH4-R23	Every load moment is a polynomial $Q_p(I)$ of stated degree, roots a, b , leading coefficient $(-1)^p p! (\lambda/\delta)^p$	EXTENDS	Equivalent moments sit in the KT PGF; the named I -economy was not found
CH5-R22	Telescoping / size-bias criterion for “burst law = QSD”	EXTENDS	KT 1982 already has the identity; this is a new <i>proof and characterisation</i>
CH5-R26–R27	$\mu = 0$: $\tau \mid \mathcal{K} = n$ harmonic / Gumbel limit	EXTENDS	Classical Yule maxima; not written as a lytic-cell assay law
CH5-R31–R34	Multi-founder kernels g_k ; $\mu = 0$ yield $k + \lambda/\delta$ (sub-extensive)	EXTENDS	Branching property is classical; this yield identity and the free-sum trap were not found
CH5-R35–R42	Chained immediate-transfer BDC: negative-binomial rupture sizes, interval Laplace transforms	EXTENDS	Textbook sums of geometrics; the laboratory was not found under this name
CH8-R03	Replicator–suppressor CTMC (birth vs unreplenished one-to-one annihilation)	EXTENDS	Pilyugin–Antia 2000 (finite killers that <i>recycle</i>); Lanchester linear law
CH8-R11–R12	$\mathcal{R}(\vartheta) = V_\infty a^{-\vartheta}$, threshold $\vartheta^* = \log V_\infty / \log a$	EXTENDS	Geometric stop-loss is actuarial folklore; not found as a BDC + integer immune budget
CH8-R17	Under threshold removal the ChPop flooding criterion <i>reverses</i> ($L < 1$ vs $L > 1$)	EXTENDS	Yuan–Allen 2011 reverse budding vs bursting under a mass-action immune term (different mechanism); Komarova flooding is unconditional

2.2 Real extensions (ancestor plus increment)

ID	Short name	Class	Nearest literature
CH3- R14, R17	$A(p) = 2\psi_{2p}(1/2)$; hypertranscendence of the logistic Koenigs function as a corollary	EXTENDS	Kolmogorov 1938 <i>A</i> ; Koenigs 1884; Becker–Bergweiler 1995
CH3- R23, R24, R26	Binary product remainder, hybrid scheme, negative elementary-form search	EXTENDS	Pollak 1969, Seneta 1968, Pakes 2026; Imomov’s candidate fails on discrete binary
CH5- R23– R24, R29	Three conditional burst means; late-burst size-bias	EXTENDS	Geometric QSD (KT). Assay: late bursts are larger
CH5- R45–R48	Single-cell BDC vs budding table; <i>Francisella</i> <i>tularensis</i> / anthrax numerical rows	EXTENDS	Pearson 2011 population comparison with <i>fixed N</i>
CH6- R04–R06	Nelson’s age-structured system forced by closed-form BDC kernels (no free $P(a)$)	EXTENDS	Nelson et al. 2004; McKendrick–von Foerster
CH6- R09–R10	Exponential-phase map $(p_{\text{eff}}(r), d_{I,\text{eff}}(r))$ with BDC endpoints	EXTENDS	Euler–Lotka; Diekmann; Wallinga–Lipsitch. Fitted p is a phase property
CH6- R16	Concrete 3-to-2 identifiability obstruction	EXTENDS	Nelson 2004; Miao et al. 2011
CH6- R22–R24	Growth–extinction tradeoff / Pareto in δ (numerical)	EXTENDS	Pearson slower burst establishment; Komarova budding spreads faster
CH7- R18	$\lambda_2 = 0$ ordinary Bessel mix	EXTENDS	AK modified-Bessel bi-critical (different ODE)
CH8- R05–R06	Certainty diagonal $\pi_{j,j}$; rate-independent $i > j \Rightarrow$ escape	EXTENDS	Pilyugin’s P^* depends on rates because killers recycle
CH8- R18	Distributional flooding vs batch-count flooding	EXTENDS	Komarova is the batch-count argument
CH9- R03	Two-layer unused- <i>individual</i> -capacity clock on a Yule process	EXTENDS	Kendall 1948 TD-Yule; Sepkoski; Rabosky/Etienne DDD on <i>species</i> count
CH10- R06, R08	BDC macrophage burst $\rightarrow$ slightly supercritical BD $\rightarrow$ host-to-host GW	EXTENDS	Antia 2003; Pearson; Gilchrist nested WH–BH; Kendall
CH2- R24	Gated release with opening intensity ηA_t	EXTENDS	Davis PDMP; Anick–Mitra–Sondhi on/off

2.3 What not to claim as new

These are the statements the chapters (or Chapter 1’s roadmap) are most likely to over-claim. Cite them as previous work and move the emphasis to the increment above.

ID	Short name	Class	Nearest literature
—	The BDC process with cemetery H	ALREADY	Karlin and Tavaré 1982. Brockwell–Gani–Resnick 1982 is the <i>more general</i> neighbour (immigration, random catastrophe sizes)
—	Closed forms $I(t)$, $D(t)$, $\hat{I}$, $J(t)$, $K(t)$, roots a, b	ALREADY	Karlin–Tavaré 1982 linear-fractional PGF
—	Occupancy geometric at every t ; QSD geometric; burst size = QSD	ALREADY	KT 1982 (2.8)–(2.9); Zhang–Zhu 2011 uniqueness. Chapter 5 headlines are 1982; the telescoping <i>proof</i> is the increment
—	Mean productive lifetime $\lambda^{-1} \log(a/(a-1))$	ALREADY	KT Green function
—	$A(p) = \lim S_n/(2p)^n$; $S_n \sim 2/n$ at criticality; Yaglom mean $1/A$	ALREADY	Kolmogorov 1938; Yaglom 1947. Kolmogorov is never cited in Chapter 3
—	Constant- p BMVR	ALREADY	Perelson 1996; Nowak–Bangham 1996; McLean 1993
—	R_0 unchanged by bursting vs dripping at matched lifetime yield	ALREADY	Pearson 2011, opening paragraph
—	Age-structured replacement of pI	ALREADY	Nelson et al. 2004 (as a <i>system</i>). The increment is BDC kernels, not the renewal skeleton
—	Comparing bursting and budding	ALREADY	Pearson 2011; Komarova 2007; Yuan–Allen 2011
—	Flooding as a biological idea	ALREADY	Komarova 2007 (replace the flooding placeholder cite)
—	Finite immune budget / inoculum escapes killing	ALREADY	Pilyugin–Antia 2000; Antia–Koella 1994; inoculum effect; complement consumption (as a <i>mechanism</i>)
—	“Nobody has published a logistic speciation model”	False	Sepkoski; Rabosky DDL; Etienne–Haegeman DDD; Kendall TD-Yule
—	A no-death Yule tests Nee’s push of the past	Cannot	POTPa needs extinction and conditioning on survival
—	Neutrophil-fraction threshold as the evolved <i>Yersinia pestis</i> switch	ALREADY	Ke, Chen and Yang 2013: switch is temperature / pH / YCV, not p_N . The displayed formula also has a sign error
—	Chapter 1: second moment K “beyond the current literature”	Not sustained	Equivalent to $\partial_{zz}G(1, t)$ from KT 1982

continued

ID	Short name	Class	Nearest literature
—	Chapter 2 hypergeometric integral identity	ALREADY	Gradshteyn–Ryzhik 3.194.1 / DLMF 15.6.1
—	Chapter 2 ABD generating function as “a result of this thesis”	ALREADY	Kendall interior PGF integrated against exponential absorption

2.4 Counts (reconciled)

Chapter	Items	Original	Extends	Already	Not a result
1 Introduction	26	0	1	14	11
2 Mathematical introduction	58	0	7	47	4
3 Galton–Watson and $A(p)$	28	0	5	22	1
4 BDC core	42	0	5	28	9
5 BDC distribution	49	0	26	22	1
6 Renewal / population	33	2	23	5	3
7 Two-type BDC	31	2	13	8	8
8 Bursting and budding pathogenesis	21	0	13	5	3
9 Variability in evolution	17	0	9	5	3
10 Conclusion, <i>Yersinia pestis</i>	15	0	7	6	2
Total	320	4	109	162	45

2.5 Citations the thesis is missing or under-using

Add these where the corresponding result is discussed. They are the difference between “we derived $I(t)$ ” and “Karlin–Tavaré gave $I(t)$; we add W_t and the I -polynomials.”

- **Karlin and Tavaré (1982)**, *Linear birth and death processes with killing*, J. Appl. Prob. 19:477–487 — the missing citation for Chapters 4 and 5.
- **Zhang and Zhu (2011)** — uniqueness of the geometric QSD under dual absorption.
- **Kolmogorov (1938)** — Chapter 3’s $A(p)$ and $S_n \sim 2/n$.
- **Yaglom (1947)**; Harris (1963); Athreya–Ney (1972).
- **Pearson, Krapivsky and Perelson (2011)** — already cited; must be credited with R_0 invariance and the establishment comparison, not only “the question.”
- **Yuan and Allen (2011)** — must be cited in Chapter 8. Published ranking-switch of budding vs bursting once an immune term is present.
- **Komarova (2007)** — replace the flooding placeholder.
- **Nelson, Gilchrist, Coombs, Hyman and Perelson (2004)** — the renewal/age-structured system Chapter 6 specialises.
- **Miao, Xia, Perelson and Wu (2011)** — structural non-identifiability.
- **Pilyugin and Antia (2000)**; **Antia and Koella (1994)** — finite nonspecific killers (they recycle; that is why their threshold depends on rates).

- **Nee, May and Harvey (1994); Rabosky and Lovette (2008); Etienne and Haegeman (2012); Pannetier et al. (2021)** — Chapter 9 cannot claim uniqueness of a variable-rate alternative to POTPa.
- **Ke, Chen and Yang (2013); Cavanaugh and Randall (1959)** — the *Yersinia pestis* switch is temperature/F1/Yop, not a neutrophil fraction.
- **Pakes (2026)** on the Kolmogorov constant — no known explicit formula; Imomov’s candidate fails on discrete binary.

2.6 Suggested sentence shapes

Do not write “we obtain the no-rupture probability $I(t)$.” Write:

Karlin and Tavaré (1982) already give the linear-fractional generating function of the linear killing process, and with it $I(t)$, the geometric occupancy, and the geometric killing position. What is developed here is the finite-time dump W_t and the organisation of every load moment as a polynomial in I . Those are the kernels that later chapters feed into a within-host renewal system.

Do not write “we compare bursting and budding.” Write:

Pearson et al. (2011) showed that at matched lifetime yield the mean-field kinetics coincide while stochastic extinction need not. Substituting the BDC burst law yields the identity $\Delta z_{\text{ext}} = (L - 1)(1/m - 1)$, which Pearson’s fixed- N comparison cannot flip. Under a finite immune toll the same L criterion reverses (cf. Yuan and Allen 2011 for a different immune mechanism).

Do not write “we solve the two-type BDC.” Write:

Antal and Krapivsky (2011) solved the triangular two-type birth–death process for terminal counts. Here the same Gauss reduction is applied to the no-catastrophe probabilities under type-specific linear killing. The objects differ: setting $\delta_i = 0$ in S, G does not recover their generating functions; it recovers them from the killed PGF.

Chapter 3

Chapter 1 — Introduction

Source. CH1/document MAIN/sections/. 26 items. No numbered theorems.

Scope. Frames early *Yersinia pestis* infection as a stochastic small-count problem; reviews branching and introduces BDC; sketches biphasic plague pathogenesis, BMVR, bursting vs budding (including Komarova flooding); states four research questions; roadmaps later chapters.

Verdict. This chapter *presents* results that later chapters derive. Almost nothing here is original. Its job, for novelty, is to name the ancestors correctly so later claims of extension have a place to sit.

Emphasize (weakly). CH1-R10 is not new mathematics (Kendall/Haldane/Antia: establishment near criticality is sensitive to founder number). As a *reading* of the macrophage phase it is the biological motivation for the BDC cell. Cite Antia et al. (2003) and Cavanaugh/Ke; do not call the branching identity a result of the thesis. The four research questions (R18–R21) are the right organisers; each already has a literature.

Do not claim. Agricultural peak-shift (R01) as a result (the ~ 6000 -year split is published; the saddle-crossing gloss is an uncited hypothesis). BDC as a process introduced here (R07). Flooding (R14; Komarova 2007). BMVR (R12). Biphasic pathogenesis (R08–R09; Cavanaugh 1959; Pujol–Bliska; Ke 2013). The roadmap’s remark that the second moment of BDC “extends beyond the current literature” (R23) is not sustained: K is in the Karlin–Tavaré PGF.

ID	Short name	Class	Nearest literature
R01	Agricultural emergence / peak-shift	EXTENDS (hypothesis)	Achtman 1999; Rasmussen 2015; Rascovan 2019; Newman–Cohen–Kipnis 1985
R02	Early-infection stochasticity	ALREADY	Standard small-count branching / epidemic establishment
R03	Brownian encounters	NOT A RESULT	—
R04	GW / Yule / BD recap	ALREADY	Galton–Watson; Yule; Kendall
R05	Critical / super / sub	ALREADY	Athreya–Ney
R06	Near-critical sensitivity	ALREADY	Kendall $(\mu/\lambda)^n$; Haldane
R07	BDC introduced	ALREADY	Brockwell–Gani–Resnick 1982; Karlin–Tavaré 1982

continued

ID	Short name	Class	Nearest literature
R08	Biphasic YP pathogenesis	ALREADY	Cavanaugh and Randall 1959; Pujol and Bliska 2005; Ke et al. 2013
R09	F1 / Yop / Pla	ALREADY	Du 2002; Sebbane 2006; Ke 2013
R10	Macrophage as near-critical founder boost	ALREADY (as maths)	Antia et al. 2003; Hataye 2019; Kendall
R11	HIV as comparator	ALREADY	Perelson 1996; Hataye 2019 intermittency
R12	BMVR	ALREADY	Nowak 1996; Perelson 1996; McLean 1993
R13	Bursting vs budding	ALREADY	Pearson 2011; Komarova 2007; Yuan–Allen 2011
R14	Flooding advantage	ALREADY	Komarova 2007. Must cite here
R15	Eclipse / multiscale aside	ALREADY	Nelson 2004; Rong; Guedj; Ciupe 2024. Sets up Ch. 6
R16	Punctuated equilibria motivation	NOT A RESULT	Eldredge–Gould
R17	Push of the past	ALREADY	Nee et al. 1994
R18	RQ1: why biphasic YP	NOT A RESULT	Question
R19	RQ2: bursting advantage	NOT A RESULT	Question
R20	RQ3: BMVR intracellular update	NOT A RESULT	Question
R21	RQ4: variable-rate evolution	NOT A RESULT	Question
R22	Roadmap Ch. 2	NOT A RESULT	—
R23	Roadmap BDC “new second moment”	ALREADY (claim false)	Karlin–Tavaré. Over-claim. Fix.
R24	Roadmap two-type / chained	NOT A RESULT	Points at Ch. 5/7
R25	Roadmap bursting–budding models	NOT A RESULT	Points at Ch. 6/8
R26	Roadmap evolution / conclusion	NOT A RESULT	Points at Ch. 9/10

Why it matters. It is the only place the four questions sit together. For novelty, its contribution is attribution: every later original statement needs the ancestor named here so the increment is visible.

Chapter 4

Chapter 2 — Mathematical introduction

Source. CH2/sections/ plus appendices A–D. 58 items.

Scope. Methods: discrete and continuous Markov chains, branching, linear birth–death, BDC as Def. 2.4, method of characteristics, small-population examples, absorption models, a hypergeometric integral, coefficient extraction. The chapter flags two items as “of this thesis”: the absorption–birth–death generating function and the hypergeometric identity. Neither is original.

Verdict. No original statements. This is a toolkit chapter. Do not let worked examples (gated release, log-spectral mean, ABD) drift into “we introduce the process.”

Emphasize. Nothing as new mathematics. Optionally, as worked models: gated release (R24), opening intensity ηA_t , sits in Davis’s PDMP class; published on/off gates are autonomous. Log-spectral / logistic-speciation mean (R18, R20–R22) is Kendall time-inhomogeneous Yule with an individual-capacity clock; a preview of Chapter 9, not DDD.

Do not claim. ABD generating function (R49) as a result of this thesis (Kendall interior PGF integrated against exponential absorption, rewritten by GR 3.194.1). Hypergeometric identity (R56) as original (GR 3.194.1 / DLMF 15.6.1). BDC (R12) and W_t (R13) as a new process (Karlin–Tavaré 1982).

ID	Short name	Class	Nearest literature
R01	Chapman–Kolmogorov	ALREADY	Norris; Karlin–Taylor; Allen
R02	Simple random-walk law	ALREADY	Feller; Karlin–Taylor
R03	Birth–death embedded walk	ALREADY	Standard embedding
R04	Gambler’s ruin	ALREADY	Feller
R05	Galton–Watson process	ALREADY	Harris; Athreya–Ney
R06	Binary GW survival iteration	ALREADY	Textbook
R07	Competing clocks	ALREADY	Exponential races
R08	Gillespie direct method	ALREADY	Gillespie 1977
R09	Poisson process law	ALREADY	Textbook
R10	Yule process law	ALREADY	Yule; Kendall

continued

ID	Short name	Class	Nearest literature
R11	Linear birth–death process	ALREADY	Kendall 1948
R12	Birth–death–catastrophe process	ALREADY	Karlin–Tavaré 1982 linear killing. Chapter over-claims if read as new
R13	Released count and burst size	ALREADY	KT Lemma 1, killing position $K = X(T_H)$
R14	Non-catastrophe factorisation	ALREADY	Branching property
R15	Catastrophe is not mean-field	ALREADY	Jensen / convexity caution
R16	Inhomogeneous Kolmogorov	ALREADY	Standard
R17	Inhomogeneous linear BD mean and extinction	ALREADY	Kendall
R18	Seasonal Lotka–Volterra process	EXTENDS	Classical LV as illustration
R19	Seasonal LV simulation	NOT A RESULT	Figure
R20	Logistic-speciation model	EXTENDS	Kendall TD-Yule; preview of Ch. 9
R21	Log-spectral mean closed form	EXTENDS	Kendall $\mathbb{E} = \exp \int \beta$; explicit limit $(C/N_0)^{\sigma C/r}$
R22	Logistic BD variant / turnover	EXTENDS	Same clock
R23	ODE–CTMC couplings and PDMP generator	ALREADY	Davis 1993
R24	Gated release process ηA_t	EXTENDS	Davis PDMP; Anick–Mitra–Sondhi autonomous on/off
R25	Gated balance and non-closed mean	ALREADY	
R26	Surging gated release	ALREADY	
R27	Gated vs surging ensemble	NOT A RESULT	Figure
R28	Coupled-system simulation	ALREADY	
R29	First-step principle	ALREADY	Feller
R30	Branching extinction fixed point	ALREADY	Harris
R31	Forward/backward master	ALREADY	Kolmogorov
R32	Birth–death PGF PDE	ALREADY	Kendall
R33	Kendall birth–death generating function	ALREADY	Kendall 1948
R34	Birth–death mean	ALREADY	
R35	Birth–death variance	ALREADY	
R36	Critical infinite mean absorption time	ALREADY	

continued

ID	Short name	Class	Nearest literature
R37	Birth–death extinction probability	ALREADY	$(\mu/\lambda)^n$
R38	Subcritical survival amplitude	ALREADY	Continuous-time Kolmogorov
R39	CT conditional mean and QS limit	ALREADY	Yaglom / Kendall
R40	Continuous-time amplitude in p	ALREADY	
R41	Discrete amplitude and two constants	EXTENDS	Comparison of two classical survival amplitudes, not a new constant
R42	Method of characteristics	ALREADY	PDE method
R43	Absorption–death generating function	ALREADY	Linear PGF
R44	Early-generation survival near criticality	ALREADY	
R45	k -founder survival and establishment	ALREADY	Branching property
R46	Cohort longevity / criticality sensitivity	NOT A RESULT	Reading
R47	Absorption-only binomial PGF	ALREADY	Binomial
R48	ABD model and Riccati characteristic	EXTENDS	Two-type CTBP degeneration (Kendall interior + absorption)
R49	ABD generating function	ALREADY	Kendall + exponential absorption + GR 3.194.1. Chapter over-claims
R50	ABD generating-function checks	NOT A RESULT	Internal
R51	ABD convolution representation	ALREADY	Standard first-event decomposition
R52	ψ hypergeometric form	ALREADY	GR / DLMF
R53	ABD domain of validity	ALREADY	Domain of R49
R54	ABD resonance	ALREADY	
R55	ABD coefficient recovery	ALREADY	
R56	Hypergeometric integral identity	ALREADY	GR 3.194.1 / DLMF 15.6.1. Not a result of this thesis
R57	Coefficient extraction (Maclaurin / Cauchy)	ALREADY	Standard
R58	Multilinear trinomial probabilities	ALREADY	Multinomial theorem

Why it matters. It supplies the notation and the BDC definition later chapters use. Novelty for the thesis does not live here. The danger is the opposite: presenting standard material, or table integrals, as derived results.

Chapter 5

Chapter 3 — Galton–Watson and $A(p)$

Source. CH3/chapterB.tex. 28 items.

Scope. Binary Galton–Watson ($p_2 = p$, $p_0 = 1 - p$). Survival recursion; subcritical Yaglom law with mean $1/A(p)$; existence of $A(p) = \lim S_n/(2p)^n$; identification with the Koenigs function of the logistic map; Becker–Bergweiler hypertranscendence; critical $S_n \sim 2/n$; a hybrid scheme for computing $A(p)$; a negative search for an elementary formula.

Verdict. No original statements. Kolmogorov is never cited, and most of the chapter is Kolmogorov (1938) / Yaglom (1947) / Harris. What to emphasize is a dictionary and a computational increment, plus a useful negative experiment (Imomov’s formula fails on discrete binary).

Emphasize.

1. $A(p) = 2\psi_{2p}(1/2)$ (R14). Koenigs 1884 linearisation of the logistic map, evaluated at $1/2$, is Kolmogorov’s constant. Not a new object: a dictionary. Matters because a closed form for A would require a closed form for ψ .
2. Hypertranscendence of $z \mapsto \psi_r(z)$ (R17). Corollary of Becker–Bergweiler 1995. Closes the “formula for ψ , hence for A ” route. Does *not* prove that $p \mapsto A(p)$ is non-elementary.
3. Truncation bound and hybrid scheme (R23–R24). Pollak 1969 / Seneta 1968 / Pakes 2026 already iterate $\mu_n = m^n/S_n$. The binary product remainder and the $2-4p$ switch are the increment.
4. Negative elementary-form search (R26). Pakes (2026): no known explicit expression. Imomov–Murtazaev’s candidate holds for linear-fractional offspring and for continuous-time quadratic, and *fails* for discrete binary (at $p = 0.3$: 2.25 vs 2.825). The CT cousin *is* elementary; binary $A(p)$ is not known to be.

Do not claim. Existence of $A(p)$ (R07, R11) — Kolmogorov 1938. Yaglom mean $1/A(p)$ (R08–R09). Critical $S_n \sim 2/n$ (R19–R21). That the chapter “introduces” $A(p)$. **Fix:** cite Kolmogorov (1938) in the first paragraph that defines $A(p)$.

ID	Short name	Class	Nearest literature
R01	Binary GW model	ALREADY	Galton–Watson; Harris; Athreya–Ney
R02	Unconditional mean ($2p$) ⁿ ; regimes	ALREADY	Textbook

continued

ID	Short name	Class	Nearest literature
R03	Offspring PGF; extinction recursion	ALREADY	Harris
R04	Survival recursion	ALREADY	
R05	Ultimate survival / extinction certainty	ALREADY	$p \leq 1/2 \Rightarrow \eta = 1$
R06	Finite- n conditional mean	ALREADY	
R07	$A(p)$ definition, $S_n \sim A(p)(2p)^n$	ALREADY	Kolmogorov 1938
R08	QS mean $1/A(p)$	ALREADY	Yaglom
R09	Yaglom-limit existence	ALREADY	Yaglom 1947
R10	Limiting conditional variance	ALREADY	Harris / Athreya–Ney
R11	Infinite-product existence, bounds	ALREADY	Kolmogorov; telescoping
R12	Logistic rescaling of survival	ALREADY	Logistic map $x \mapsto 2px(1-x)$ is the binary GW PGF
R13	Koenigs linearisation	ALREADY	Koenigs 1884
R14	$A(p) = 2\psi_{2p}(1/2)$	EXTENDS	Dictionary Kolmogorov $\leftrightarrow$ Koenigs
R15	BB hypertranscendence of the inverse	ALREADY	Becker–Bergweiler 1995
R16	Inverse preserves DA	ALREADY	Standard lemma (Fernandes 2021)
R17	HT of the logistic Koenigs function	EXTENDS	Corollary of R15–R16
R18	Elementary Koenigs at $r = 2, 4$	ALREADY	Classical
R19	Critical $S_n \sim 2/n$	ALREADY	Kolmogorov 1938
R20	Infinite expected lifetime at criticality	ALREADY	
R21	Critical power-law tails	ALREADY	
R22	Near-critical $A \sim 2\varepsilon$	ALREADY	Kolmogorov amplitude
R23	Truncation bound / stopping rule	EXTENDS	Pollak / Seneta / Pakes
R24	Product–asymptotic hybrid	EXTENDS	Same line; binary remainder is the increment
R25	Complex-plane Koenigs geometry	NOT A RESULT	Visual
R26	Elementary-formula search (negative)	EXTENDS	Pakes 2026; Imomov fails on binary
R27	Unconditional factorial / second moments	ALREADY	Harris
R28	Near-critical moment rates	ALREADY	Yaglom + Kolmogorov amplitude

Why it matters. Binary GW is the discrete counterpart of linear birth–death, and $A(p)$ is the discrete cousin of the Karlin–Tavaré root b . The chapter’s genuine service is to show why the discrete constant is harder than the CT one, and to give a working numerical scheme.

Chapter 6

Chapter 4 — BDC: definition and main results

Source. CH4_REWRITE_RESTORED/sections/. 42 items.

Scope. Linear BDC as one lytic infected cell: per-capita rates $\lambda n, \mu n, \delta n$, cemetery H , dump coordinate W_t . Riccati for $I(t)$; roots a, b ; $I, D, \hat{I}$; load moments J, K ; release moments of W_t ; burst-size first two moments; I -polynomial organisation.

Verdict. This is the chapter most at risk of over-claiming. Karlin and Tavaré (1982) already contain the process, the Riccati / linear-fractional PGF, $I(t)$, the geometric killing position with mean $a/(a-1)$, and the geometric QSD. Dictionary: their (a, b, c) (death, birth, killing) are this chapter's (μ, λ, δ) . What is *not* in that paper, and should be emphasized, is the finite-time dump calculus and the I -economy.

Emphasize.

1. Dump process W_t (R17, R18, R20, R22). KT give the *eventual* killing position $K = X(T_H)$. The chapter keeps a time-dependent released count $W_t = K \mathbf{1}_{\{t \geq \tau\}}$ and closes $V(t)$, $\mathbb{E}[W_t^2]$, $\text{Var}(W_t)$, and current yield as functions of $I(t)$. These were not found. They matter because Chapter 6's age kernels are $\delta K(\alpha)$ and $V(\alpha)$.
2. Load moments as polynomials in I (R23). Equivalent moments live in the PGF. The named proposition (degree $p+1$, roots at a and b , leading coefficient $(-1)^p p! (\lambda/\delta)^p$) was not found. Every later kernel is then a polynomial in one monotone clock.

Cite as known (KT 1982). Process with cemetery H (generator (0.4), §2); roots a, b (2.2); linear-fractional PGF (2.5); $I(t) = P(T_H > t)$ (2.7); geometric killing position, mean $a/(a-1)$ (2.8); geometric QSD (2.9), uniqueness by Zhang–Zhu 2011. Also already: J and K as derivatives of the PGF (so Chapter 1's “second moment beyond the literature” is false). Brockwell–Gani–Resnick 1982 is the right historical citation but is *more general*.

ID	Short name	Class	Nearest literature
R01	Linear BDC with release	ALREADY	KT 1982; Brockwell et al. 1982 more general
R02	Killing vs catastrophe (H vs 0)	ALREADY	KT already keep H ; Baake et al. 2024 revisit the naming

continued

ID	Short name	Class	Nearest literature
R03	Riccati for I	ALREADY	KT; linear BD Riccati (Kendall)
R04	Characteristic roots a, b	ALREADY	KT (2.2)
R05	Vieta / gap identities	ALREADY	Elementary from the quadratic
R06	$I(t)$ closed form	ALREADY	KT (2.7)
R07	$b =$ extinction-before-rupture	ALREADY	KT $P(\text{hit } 0 \text{ before } H)$
R08	$D(t)$ internal extinction	ALREADY	Split of the PGF at $z = 0$
R09	$\hat{I} = I - D$	ALREADY	KT two conditionings
R10	$\hat{I}$ initial slope	ALREADY	Differentiation of R09
R11	Multi-founder $I^k - D^k$	ALREADY	Branching property
R12	Non-fixation does not factorise	ALREADY	Same
R13	$\dot{I} = -\delta J$	ALREADY	Flux / Green
R14	Mean load J	ALREADY	$\partial_z G(1, t)$
R15	Open J - K hierarchy	ALREADY	Standard moment ODEs
R16	Second moment K	ALREADY	$\partial_{zz} G$; Chapter 1 over-claim
R17	Mean release $V(t)$	EXTENDS	Eventual V_∞ is KT; finite-time $V(t)$ not found
R18	Release second moment	EXTENDS	Eventual geometric second moment is KT
R19	$\text{Var}(X)$	ALREADY	$K - J^2$
R20	$\text{Var}(W)$	EXTENDS	Finite-time dump variance not found
R21	Exclusion / $\text{Cov}(X, W) = -JV$	NOT A RESULT	Definitional ($XW \equiv 0$)
R22	Current yield $\mathbb{E}[X + W]$	EXTENDS	Bookkeeping in the W calculus
R23	Load moments polynomials in I	EXTENDS	Organisation of the PGF; named proposition not found
R24	I as clock	NOT A RESULT	Time change of an autonomous ODE
R25	Burst-size first two moments	ALREADY	KT geometric, mean $a/(a - 1)$
R26	Geometric burst preview, $\mu = 0$	ALREADY	Yule + killing; KT
R27	Burst-time density δJ	ALREADY	Flux
R28	Productive conditional mean plateau	ALREADY	QSD mean (KT / Zhang-Zhu)
R29	$J/\hat{I}$ vs J/I	ALREADY	Two conditionings
R30	No-death limit	ALREADY	
R31	No-catastrophe $\rightarrow$ BD	ALREADY	Kendall
R32	Critical births = deaths	ALREADY	
R33	No-births limit	ALREADY	
R34	Four-ODE reconstruction	NOT A RESULT	Internal check
R35	Landscape of b and V_∞	NOT A RESULT	Figure

continued

ID	Short name	Class	Nearest literature
R36	Mean load mixes fates	NOT A RESULT	Reading of J
R37	Load SD exceeds mean	NOT A RESULT	Reading of R19
R38	Endpoints	NOT A RESULT	Consistency
R39	Special case of Brockwell	ALREADY	Correct placement
R40	$I_{\pm}$ notation	NOT A RESULT	Dictionary
R41	Non-explosion	ALREADY	Linear rates; Cairns–Pollett Thm 1
R42	Three research questions	NOT A RESULT	Framing

Suggested wording.

The linear killing process of Karlin and Tavaré (1982) is the intracellular model. Their generating function already supplies I , J , K , and the eventual burst-size moments. The present chapter adds the finite-time released count W_t and shows that every load moment is a polynomial in I . Those two additions are what the population chapter consumes.

Why it matters biologically. A lytic cell (*Yersinia pestis* in a macrophage, *Francisella tularensis*, lytic phage) does not drip. Per-capita catastrophe δn makes a crowded cell more likely to burst. Distinguishing internal clearance (D) from rupture ($1 - I$) is the difference between an empty intact cell and a dump. $V(t)$ is the expected dump by age t ; that is the object a within-host model needs, not a fitted constant p .

Chapter 7

Chapter 5 — BDC distribution, QSD, burst statistics

Source. CH5_REWRITE/sections/. 49 items.

Scope. Complete law of one bursting cell: generating function, sliding geometric occupancy, QSD, burst-size identity, burst time, conditional means, several founders, chained transfer, single-cell bursting vs budding. The chapter organises itself around one identity: the burst-size law equals the quasi-stationary law.

Verdict. That identity is Karlin and Tavaré (1982), eqs. (2.8)–(2.9). Zhang and Zhu (2011) restate uniqueness of the geometric QSD. Kendall (1948) already has the unkilld sliding geometric. Appendix rows marked N for $P(t)$, p_n , QSD, burst law, and $\mathbb{E}[T_{\text{prod}}]$ should be S (standard) or R (recap of Chapter 4). What to emphasize is the proof, the timing laws, the multi-founder yield, and the chained-transfer laboratory.

Emphasize.

1. Telescoping / size-bias criterion (R22). New *proof* that burst = QSD, and a characterisation (linear hazard + sliding geometric + $p_1 = P'/\lambda$).
2. $\mu = 0$ harmonic / Gumbel timing (R26–R27). $\tau \mid \mathcal{K} = n$ as a sum of exponentials; Gumbel limit. Budding comparator is linear in n .
3. Multi-founder kernels (R31–R34). g_k ; free-sum trap; $\mu = 0$ lifetime yield $k + \lambda/\delta$ (sub-extensive unconditionally). Several founders sharing one cell do not add solo yields.
4. Chained immediate transfer (R35–R42). Wholesale transfer of the load. Negative-binomial rupture sizes are textbook sums of geometrics; the laboratory (and fragility as soon as $\mu > 0$) was not found under this name.
5. Three conditional burst means / late-burst factor (R23–R24, R29). Size-bias of the geometric QSD. “Factor of two” is $a \rightarrow 1$, not universal.
6. Single-cell BDC vs budding (R45–R46) and *Francisella tularensis*/anthrax rows (R48). Pearson 2011 already has mean-matched burst vs continuous at population level with fixed N . Numerical timings at Carruthers *Francisella tularensis* and Williams anthrax rates are new evaluations.

Do not claim as new. Sliding geometric occupancy (R08, R11, R12) — KT (2.5). QSD geometric (R14) — KT (2.9). Burst size = QSD (R20, R21) — KT (2.8), the headline. Mean productive lifetime (R18) — KT Green function $\lambda^{-1} \log(a/(a-1))$. Recap of Chapter 4 closed forms.

ID	Short name	Class	Nearest literature
R01	BDC process	ALREADY	KT 1982
R02	Joint process / W_t	ALREADY	KT killing position; Ch. 4
R03	Roots, AB identity	ALREADY	Ch. 4 / KT
R04	$I, D, \hat{I}$	ALREADY	Ch. 4 / KT
R05	J, K, V , conditional mean burst	ALREADY	Ch. 4 / KT
R06	Second moment of one-cell release	ALREADY	Ch. 4 (eventual); finite-time is CH4-R18
R07	Kolmogorov equations, PGF PDEs	ALREADY	
R08	Single-founder GF	ALREADY	KT (2.5) linear-fractional
R09	k -founder GF	ALREADY	Branching property
R10	Two rupture conventions	EXTENDS	CH4-R02; bookkeeping
R11	State probabilities	ALREADY	Coefficients of (2.5)
R12	Load is geometric at every time	ALREADY	KT; Kendall unkilld
R13	Positive states carry $\hat{I}$	ALREADY	
R14	QSD	ALREADY	KT (2.9); Zhang–Zhu 2011
R15	QSD moments	ALREADY	Geometric moments
R16	QSD decay rate	ALREADY	
R17	Decay rate = BDC time constant	ALREADY	
R18	Mean productive lifetime	ALREADY	KT Green
R19	Occupation integral of p_k	ALREADY	Green function
R20	Burst-size distribution	ALREADY	KT (2.8)
R21	Burst size = QSD	ALREADY	KT identity
R22	Telescoping criterion	EXTENDS	New proof / characterisation of R21
R23	Size-biased burst given rupture time	EXTENDS	Geometric size-bias
R24	Late-burst mean	EXTENDS	
R25	Defective burst-time density	ALREADY	Flux δJ
R26	$\tau \mid k, \mu = 0$	EXTENDS	Yule / exponential order stats
R27	Conditional CDF, Gumbel limit	EXTENDS	
R28	Mean burst time on bursting	EXTENDS	Evaluation
R29	Three conditional burst means	EXTENDS	Circularity warning
R30	k -founder fixation	ALREADY	Branching property
R31	k -founder moments, g_k	EXTENDS	
R32	Superlinear flux, free-sum trap	EXTENDS	

continued

ID	Short name	Class	Nearest literature
R33	Lifetime yield from k founders	EXTENDS	
R34	$\mu = 0$ yield, sub-extensivity	EXTENDS	$k + \lambda/\delta$
R35	Chained immediate-transfer model	EXTENDS	Not found as this laboratory
R36	Event types ignore the load	ALREADY	Competing exponentials
R37	NB rupture-size law	EXTENDS	Textbook NB applied here
R38	Chained rupture-size moments / PGF	EXTENDS	
R39	Chained rupture times	EXTENDS	
R40	Inter-rupture Laplace	EXTENDS	Phase-type
R41	Mean inter-rupture interval	EXTENDS	
R42	Second rupture-time Laplace	EXTENDS	
R43	Coefficient extraction, general μ	EXTENDS	Ch. 2 method applied; $\mu > 0$ open
R44	Simulation verification	NOT A RESULT	
R45	Bursting vs budding structural table	EXTENDS	Pearson/Yuan population, fixed N
R46	Matching means $\neq$ matching support	EXTENDS	Pearson's logical point, single-cell arithmetic here
R47	Five limiting-dilution predictions	EXTENDS	Translation of R14/R21 into an assay
R48	<i>Francisella tularensis</i> / anthrax numerical extremes	EXTENDS	Carruthers 2020; Williams 2021 rates
R49	Geometric robust to k , fragile to chaining with death	EXTENDS	Synthesis

Suggested wording.

Karlin and Tavaré (1982) already proved that for linear birth–death with killing, occupancy is linear-fractional, the QSD is geometric, and the killing position is that same geometric. The present chapter gives a size-bias proof of that identity, the burst-time law at $\mu = 0$, the multi-founder yield $k + \lambda/\delta$, and a chained-transfer laboratory in which the geometric structure fails as soon as death is present. Those are the distributional facts a lytic-cell assay can test.

Why it matters biologically. A fitted mean burst size does not determine establishment (Pearson). For a genuinely lytic pathogen the whole geometric family is determined by (λ, μ, δ) , and the same three rates span *Francisella tularensis* (rare clearance, huge bursts, low ID) and anthrax (almost-sure clearance, tiny bursts, high ID). Several co-infecting founders sharing one cell do not

add solo yields.

Chapter 8

Chapter 6 — Burst-aware renewal and population comparison

Source. CH6_REWRITE/sections/. 33 items.

Scope. What replaces BMVR’s pI when release is terminal and batched. Burst-aware renewal forced by BDC kernels; classical BMVR as an exponential-phase projection; identifiability; R_0 ; flooding parameter L ; growth–extinction tradeoff.

Verdict. This chapter contains two of the four original statements in the thesis (the L -flooding identity and the $L = 1$ coincidence of laws), sitting on top of a renewal system that is Nelson et al. (2004) and an R_0 invariance that is Pearson (2011), first paragraph. Write the skeleton as an application of Nelson; write R_0 as Pearson; lead with L .

Emphasize.

1. Flooding identity (R20) — original as a statement.

$$z_{\text{ext}}^{\text{burst}} - z_{\text{ext}}^{\text{bud}} = (L - 1) \left(\frac{1}{m} - 1 \right), \quad L := a(1 - b).$$

Bursting lowers establishment extinction iff $L > 1$, i.e. $1/\delta < 1/\lambda + 1/\mu$ ($\mu = 0$ always). Pearson 2011 cannot flip this sign; Quirouette et al. 2023 already said extinction tracks the offspring *law*, not timing; Komarova 2007 floods antibodies (different object).

2. $L = 1$ coincidence of laws (R21). When $L = 1$ the two offspring laws are the same geometric. Variance order: $\text{Var}_{\text{bud}} - \text{Var}_{\text{burst}} \propto (L - 1)$.
3. Kernels from BDC, not posited (R04–R06) — extends Nelson. S and $g = \delta K$ are closed forms in (λ, μ, δ) . No free age schedule to fit. p is no longer a parameter of the organism.
4. Exponential-phase projection (R09–R10) — extends Euler–Lotka. Fitted p is a *phase* property. Do not claim the projection idea is new.
5. Identifiability 3-to-2 (R16) — extends Nelson/Miao.
6. Growth–extinction tradeoff (R22–R24) — numerical. General $r_{\text{bud}} > r_{\text{burst}}$ is unproved — leave it unproved.

Do not claim as new. BMVR (R01). Exponential kernels $\equiv$ BMVR (R07). R_0 **invariance at matched yield (R14).** **Pearson 2011 opening paragraph.** Use it, cite it, do not number it as a new theorem.

ID	Short name	Class	Nearest literature
R01	Classical BMVR / budding cell	ALREADY	Perelson 1996; Nowak 1996; McLean 1993
R02	Synchronous cohort is scheduled	EXTENDS	Obstruction to constant p
R03	Obstruction to constant p	EXTENDS	
R04	Survival and release kernels	EXTENDS	Nelson 2004 posited $P(a)$; here closed-form BDC
R05	Burst-aware renewal system	EXTENDS	Nelson eqs. (1)
R06	Universal skeleton, any kernels	EXTENDS	Same
R07	Exponential kernels $\rightarrow$ BMVR	ALREADY	
R08	Renewal verification suite	NOT A RESULT	
R09	Exponential-phase projection	EXTENDS	Euler–Lotka / Diekmann / Nelson
R10	p_{eff} map endpoints	EXTENDS	
R11	p_{eff} decreases in r (numerical)	EXTENDS	
R12	Generation kernel, R_0 , Euler–Lotka	EXTENDS	Diekmann
R13	Overlays: what the projection loses	EXTENDS	
R14	R_0 unchanged by bursting	ALREADY	Pearson 2011
R15	Continuum R_0 vs discrete m	ALREADY	
R16	Identifiability obstruction	EXTENDS	Nelson 2004; Miao 2011
R17	Fitting-practice predictions	EXTENDS	
R18	Bursting offspring PGF / moments	EXTENDS	Pearson random-burst PGF; KT size-at-killing
R19	Bursting extinction probability	EXTENDS	Pearson root of thinned PGF
R20	Flooding identity / L criterion	ORIGINAL	Pearson (no L); Quirouette; Komarova different object
R21	$L = 1$ coincidence of laws	ORIGINAL	Linear-fractional family
R22	Budding invades faster at matched R_0	EXTENDS	Pearson 1.75 d vs 2.46 d; Komarova
R23	Generation-time mechanism	EXTENDS	Wallinga–Lipsitch
R24	Growth–extinction Pareto in δ	EXTENDS	Numerical; general order unproved

continued

ID	Short name	Class	Nearest literature
R25	Five-ODE ladder	EXTENDS	Reach of the construction
R26	BMVR as fast-export / QSS	EXTENDS	
R27	Reset-catastrophe intermediate	EXTENDS	Catalogue
R28	Partial-release mean field	EXTENDS	
R29	HIV-stage renewal and Allee	EXTENDS	Application
R30	HIV mean-linear, not exponential BDC	ALREADY	Hataye / HIV production
R31	What carries beyond BDC	NOT A RESULT	Discussion
R32	Three regimes, flooding-boundary numerics	NOT A RESULT	Figures for R20
R33	Hypergeometric Laplace of kernels	EXTENDS	Ch. 4 I -polynomials in Laplace space

Suggested wording.

Nelson et al. (2004) already replaced constant p by an age-dependent production schedule. Here that schedule is not fitted: it is the BDC flux $\delta K(\alpha)$. Pearson et al. (2011) already showed that matched lifetime yield equalises R_0 while leaving stochastic extinction free. Substituting the BDC burst law yields $\Delta z_{\text{ext}} = (L - 1)(1/m - 1)$. When $L = 1$ the two offspring laws coincide. That is the chapter's original comparison; the renewal skeleton and the R_0 lemma are inherited.

Why it matters biologically. A fitted HIV p is not a property of *Yersinia pestis*. For a lytic pathogen, release and death are the same event, the extracellular curve is a convolution against a closed-form kernel, and whether bursting helps establishment is a criterion on (λ, μ, δ) , not a slogan. The same criterion will reverse in Chapter 8 once the immune functional is a threshold rather than linear thinning.

Chapter 9

Chapter 7 — Two-type BDC

Source. CH7/Two_Type_Chapter/document MAIN/sections/. 31 items.

Scope. Triangular two-type birth–death (type 1 $\rightarrow$ type 2 conversion) plus a global absorbing catastrophe at rate $\delta_1 X + \delta_2 Y$. Observables: no-catastrophe probabilities $S(t)$ and $G(t)$, not terminal counts. Type 2 is autonomous; type 1 is a forced Riccati that linearises to Gauss hypergeometric. Biological reading: early vs adapted intracellular *Yersinia pestis*.

Verdict. This chapter has the other two original statements (exact S , and the killed PGF that recovers Antal–Krapivsky). The *method* is Antal and Krapivsky (2011). Type-2 G is one-type linear BDC (Karlin–Tavaré / Chapter 4). Write the method as inherited; write the object as original.

Verified from AK arXiv:1105.1157: their backward system is this chapter’s ODEs at $\delta_1 = \delta_2 = 0$ with PGF data x, y . Survival data $S(0) = G(0) = 1$ then force $S \equiv G \equiv 1$. So $\delta = 0$ in S, G does *not* recover AK. Recovery holds for the killed PGF. Li (2024) “two-type catastrophes” is a homonym. No published ${}_2F_1$ form for type-specific linear catastrophe $\delta_1 x + \delta_2 y$ was found.

Emphasize.

1. Exact $S(t)$ (R06) — original as a statement. Wronskian mix of a real-axis ${}_2F_1$ pair along $z_2 < 0$,

$$\hat{S}(\hat{t}) = h - \hat{q}_2 z_2(\hat{t}) \frac{K_U U'(z_2) + K_V V'(z_2)}{K_U U(z_2) + K_V V(z_2)},$$

with $U = {}_2F_1$, $V = (-z)^{1-C} {}_2F_1$. Nearest miss: AK terminal-count PGF. A lineage that grew early and died can contribute more catastrophe exposure than a small lineage that remains alive.

2. Killed PGF (R23) — original as a companion. Same mix, data x, y . AK is $\delta_i = 0$ of *this* object.
3. Method (R05) — extends AK. Cite AK 2011 and Bender–Orszag. Do not claim a new special-function theory.
4. $\lambda_2 = 0$ Bessel boundary (R18) — extends. Ordinary J_ρ, Y_ρ with $\delta_2 > 0$ is not AK’s modified-Bessel bi-critical PGF.

Do not claim. Type-2 G , $\delta_2 = 0$ type-1, $\nu = 0$ (R04, R15, R19): one-type linear BDC. Under-cited. Point at Karlin–Tavaré / Chapter 4 $I(t)$. The biological EQ/MAT/GATE/EARLY comparison as a theorem (MAT is a working hypothesis).

ID	Short name	Class	Nearest literature
R01	Two-type BDC process	EXTENDS	AK skeleton + KT/Brockwell killing
R02	Backward equations	ALREADY	Standard first-step / branching
R03	Autonomous type-2	EXTENDS	Triangular structure is AK
R04	Closed form for G	ALREADY	KT / Ch. 4 $I(t)$ (one-type BDC)
R05	Gauss hypergeometric reduction	EXTENDS	AK §§II–III
R06	Exact finite-time S	ORIGINAL	AK PGF is a different object
R07–R11	Derivation steps / branch management	NOT A RESULT	Method of R06
R12	Forever-avoidance $S \rightarrow h$	EXTENDS	
R13	$G \rightarrow r_{2,-}$	EXTENDS	
R14	Equal-rate gap (no independent form)	EXTENDS	Fair negative remark
R15	$\delta_2 = 0$ elementary Riccati	ALREADY	One-type
R16	$\lambda_1 = 0$ linear integral	ALREADY	
R17	Limits catalogue	EXTENDS	
R18	$\lambda_2 = 0$ Bessel mix	EXTENDS	Not AK's I_ν, K_ν
R19	$\nu = 0$ decoupled	ALREADY	Two independent one-type BDCs
R20–R21	Parameter dictionary / scaling	ALREADY	
R22	Killed-PGF backward system	EXTENDS	AK (10)–(11) at $\delta = 0$
R23	Killed-PGF closed form	ORIGINAL	Recovers AK at $\delta_i = 0$
R24	Ultimate counts on catastrophe-free paths	EXTENDS	
R25	Ultimate release law	EXTENDS	KT size-at-killing, two-type packaging
R26	Interpretation stub	NOT A RESULT	Empty file
R27	Biological application (four orderings)	EXTENDS	EQ/MAT/GATE/EARLY: reading, not identification
R28–R29	Companion biology extra arguments	NOT A RESULT	
R30	Early vs adapted YP phenotypes	ALREADY	Ke 2013; Pujol–Bliska
R31	MAT bursting as working hypothesis	ALREADY	Monack 1997; Peters 2013; not a theorem

Suggested wording.

Antal and Krapivsky (2011) solved the triangular two-type birth–death process for terminal population counts by reducing a Riccati system to Gauss’s equation. The present chapter keeps that triangular skeleton and that reduction, and changes the observable: a type-specific linear catastrophe clock, and the probability that it has not yet fired. Setting the catastrophe rates to zero in those survival probabilities does not recover Antal–Krapivsky; it recovers them from the killed generating function, which is the same mix with PGF data. Type 2 is the classical one-type killing probability.

Why it matters biologically. Intracellular *Yersinia pestis* is not one type. Conversion (adaptation, Yop expression) and loss of macrophage containment can run at different per-capita rates. $S(t)$ is the chance that a lineage started as the early type has not yet blown the cell, integrating exposure along the whole path — the right object if catastrophe is “being seen” rather than “being extinct.”

Chapter 10

Chapter 8 — Bursting and budding pathogenesis

Source. CH8_REWRITE/sections/. 21 items.

Scope. Does bursting help once immune capacity is depletable? Schedule-alone cannot matter (linear rates). Two nonlinear models: intracellular replicator–suppressor; extracellular fixed removal. Closed forms, a certainty threshold, $\mathcal{R}(\vartheta) = V_\infty a^{-\vartheta}$, and reversal of Chapter 6’s flooding criterion. The chapter correctly says that comparing bursting and budding is not new (Pearson, Komarova).

Verdict. No original statements in the strict sense after adversarial checking: every New mark has a mechanism neighbour (Pilyugin–Antia, geometric stop-loss, Yuan–Allen ranking-switch, Komarova flooding). The constructions and the L -reversal under a threshold functional are still the chapter’s contributions. Cite Yuan 2011 and Pilyugin–Antia 2000, which the chapter currently misses.

Emphasize.

1. Reversal of the flooding criterion (R17) — strongest extension. Same matched-mean pair as Chapter 6; functional $x \mapsto (x - \vartheta)_+$ is convex, so the Chapter 6 variance order $\propto (L - 1)$ *flips* the sign: bursting now helps when $L < 1$. Yuan and Allen (2011) already reverse budding vs bursting under an HIV-style mass-action immune term. Different mechanism, different criterion, but the qualitative published fact is the ranking-switch. The thesis contribution is the complementary L criterion on the *same pair of laws* as Chapter 6.
2. $\mathcal{R}(\vartheta) = V_\infty a^{-\vartheta}$ and $\vartheta^* = \log V_\infty / \log a$ (R11–R12). Geometric stop-loss is actuarial folklore. Not found as a BDC burst + integer immune budget.
3. Replicator–suppressor CTMC (R03) and certainty diagonal (R05–R06). Pilyugin–Antia 2000: finite macrophage pool that *recycles*; their P^* depends on rates. The consume-not-recycle choice makes $i > j \Rightarrow$ leftover replicators even with no births — combinatorial, rate-independent.
4. Two flooding arguments (R18). Batch-count is Komarova’s intuition and is elementary. Distributional (two laws, one toll) is R17.

Do not claim. Bursting vs budding as a question. Finite killing budget as a new mechanism. Schedule-alone invariance (R01) as new. YP two-phase as ϑ^* -crossing (R20): Ke 2013 answers the switch with temperature / pH / YCV. Replace the flooding placeholder cite with Komarova (2007).

ID	Short name	Class	Nearest literature
R01	Schedule-alone cannot matter	ALREADY	Pearson mean-field; Ch. 6 R_0 ; Diekmann
R02	Chained-rupture process	ALREADY	Ch. 5 laboratory; successive BDC
R03	Replicator-suppressor process	EXTENDS	Pilyugin–Antia 2000 (recycle); Lanchester
R04	Replicator-suppressor numerics	EXTENDS	
R05	Certainty threshold $i > j$	EXTENDS	Combinatorial; Pilyugin vulnerability; Li 2002 neutrophil threshold
R06	Diagonal law $\pi_{j,j}$	EXTENDS	Pochhammer product; $1/(j+1)$ at $\nu = 1$ classical
R07	Off-diagonal / simulations	EXTENDS	$\pi_{i,j}$ open in closed form
R08	Detection avoidance (named, not modelled)	NOT A RESULT	Conway–Ribeiro 2018
R09	Fixed-removal model	EXTENDS	Komarova flooding; complement consumption; integer- ϑ caricature
R10	Geometric burst law (inherited)	ALREADY	KT / Ch. 4 / Ch. 5
R11	$\mathcal{R}(\vartheta) = V_\infty a^{-\vartheta}$	EXTENDS	Geometric stop-loss
R12	$\vartheta^* = \log V_\infty / \log a$	EXTENDS	Corollary of R11
R13	Subcritical regime created by ϑ	ALREADY	$V_\infty > 1$ inherited; commentary
R14	Numerical ϑ^* rows	EXTENDS	Illustration
R15	Subductive proliferation (named, not modelled)	NOT A RESULT	
R16	Ch. 6 R_0 / L recalled	ALREADY	Ch. 6
R17	Flooding criterion <i>reverses</i>	EXTENDS	Yuan 2011 ranking-switch (different immune); convex order of Ch. 6
R18	Two flooding arguments	EXTENDS	Komarova = batch-count
R19	Selection on the bursting–budding spectrum	EXTENDS	Komarova; Wang lysis-timing; Ch. 6 spectrum
R20	YP two-phase as threshold-crossing	EXTENDS (weak)	Ke 2013 temperature cue, not ϑ^*
R21	Verification tables	NOT A RESULT	0 sign errors on reversal

Suggested wording.

Pearson, Komarova, and Yuan and Allen already compared bursting and budding. What is missing from those comparisons, and from Chapter 6’s linear-clearance flooding theorem, is a finite killing budget. Two caricatures are introduced. Intracellularly, unreplenished one-to-one annihilation (cf. Pilyugin and Antia, who recycle killers) produces a rate-independent certainty threshold. Extracellularly, an integer toll on the geometric BDC burst yields $\mathcal{R}(\vartheta) = V_\infty a^{-\vartheta}$. Because $x \mapsto (x - \vartheta)_+$ is convex, Chapter 6’s L criterion reverses. Yuan and Allen (2011) already found a ranking-switch under mass-action immunity; the present reversal is the complementary criterion on the same BDC/budding

pair.

Why it matters biologically. Macrophage granules and extracellular complement are depletable. Linear clearance (Chapter 6) cannot see a safety-in-numbers threshold. Whether bursting helps then depends on *which functional* one puts on the same two offspring laws — the point of R17–R18, and the reason Komarova’s unconditional flooding must not be copied.

Chapter 11

Chapter 9 — Variability in evolution

Source. CH9/document MAIN/sections/. 17 items. No numbered theorem environments.

Scope. High early diversification after ecological relief as a mechanistic variable-rate phenomenon, not only Nee’s push-of-the-past sampling artefact. Core: no-death time-inhomogeneous Yule whose speciation rate tracks logistic *individual* abundance. Then speculative models: Pimentel genetic feedback, rise–run–ruin–rejuvenation, birth–plague, vitality.

Verdict. No original statements. Diversity-dependent diversification (DDD) and time-dependent Yule/birth–death are a mature literature. The chapter’s claim that “nobody has published a logistic version” is false (Sepkoski; Rabosky DDL; Etienne–Haegeman DDD). A no-death Yule cannot test POTPa, which requires extinction and conditioning on survival. Pull of the present is title-only.

Emphasize (weakly).

1. Logistic-individual Yule (R03). $N(t)$ is a deterministic individual logistic; Y is an ordinary inhomogeneous Yule (Kendall 1948). DDD puts $\lambda(n) \propto 1 - n/K$ on *species* count. The two-layer unused-capacity coupling is the only residual distinction. Not a new process class.
2. Pimentel Moran + minority-resistance kernel (R08–R10). Still in the Pimentel–Levin–Pease genetic-feedback class. Simulations show reversal-to-fixation, not Pimentel’s missing oscillation. Unfinished.
3. RRRR and birth–plague (R13, R15). Assemblies of BDC+immigration and SI-with-sparking. Present as exploratory models, not results.

Do not claim. A new alternative to Nee 1994 (the last two decades of DDD vs time-dependent BD vs POTPa *are* that alternative). The explicit mean $e^{\sigma K t}((\xi + 1)/(\xi + e^{rt}))^{\sigma K/r}$ as a named discovery (three-line specialisation of Kendall). PGF / geometric occupancy (R06–R07): Kendall. Punctuated-equilibria theorem (R01).

ID	Short name	Class	Nearest literature
R01	Relaxed-selection punctuated equilibria	EXTENDS (hypothesis)	Eldredge–Gould; Newman–Cohen–Kipnis 1985; Yoder; Lahti
R02	POTPa vs variable-rate origination	ALREADY (as a question)	Nee et al. 1994; Budd–Mann 2018; DDD literature

continued

ID	Short name	Class	Nearest literature
R03	Logistic-speciation Yule	EXTENDS (barely)	Kendall TD-Yule; Sepkoski; Rabosky/Etienne DDD on species; Hubbell
R04	Ensemble slowdown figure	ALREADY (as a pattern)	Generic to declining λ ; Pannetier TD $\equiv$ DD for LTT
R05	Logistic-Yule mean	ALREADY	Kendall $\exp \int \lambda$; elementary integral
R06	Logistic-Yule PGF	ALREADY	Kendall inhomogeneous Yule PGF
R07	Occupancy distribution	ALREADY	Geometric / Kendall
R08	Pimentel	EXTENDS	Pimentel; Levin; Pease
	Moran+resistance		
R09	Reversal and divergence	EXTENDS	Reversal shown, not oscillation
R10	Critical saddle	EXTENDS	
R11	Cellular automaton	EXTENDS	Illustration of R08
R12	Pimentel + variation collapse	EXTENDS	Unfinished
R13	Rise-run-ruin- rejuvenation	EXTENDS	Assembly of known pieces
R14	RRRR cyclic realisations	NOT A RESULT	Figures
R15	Birth-plague model	EXTENDS	BDC+immigration; SI with demography
R16	Birth-plague realisations	NOT A RESULT	Figures
R17	Vitality birth-plague	NOT A RESULT	Vitality never enters the rates

Suggested wording.

Nee, May and Harvey (1994) showed that apparent early bursts can be a sampling artefact of constant-rate birth-death conditioned on survival. Diversity-dependent and time-dependent speciation models are the existing mechanistic alternatives. The model used here is a no-death Yule whose rate tracks unused *individual* capacity (Kendall 1948 with a logistic clock). It cannot adjudicate push of the past, which requires extinction. It is offered as a simple variable-rate illustration, not as a new alternative to that literature.

Why it matters biologically. The thesis's interest is whether *Yersinia pestis* (and large clades generally) can show genuinely elevated origination after ecological relief, rather than only a conditioned-BD illusion. That question is real (Budd and Mann). This chapter does not yet have a model that can answer it, because the analytic core has no death.

Chapter 12

Chapter 10 — Conclusion: evolved strategy of *Yersinia pestis*

Source. CH10/document MAIN/sections/. 15 items. No numbered theorems. (Source title: Chapter 8.)

Scope. Why *Yersinia pestis* uses a two-phase invasion: macrophage then extracellular / anti-phagocytic. Conserved-replicative-density mean invariance as a foil; two-phase within-host branching embedded in a host-to-host Galton–Watson; “quadrant” death-rate argument and a neutrophil-fraction threshold; speculative public-good accumulation inside a macrophage.

Verdict. No original statements. The biology of the switch is Cavanaugh / Pujol–Bliska / Ke 2013 (temperature, phagosomal pH, YCV — not a neutrophil fraction). The nested BDC→BD→GW chain is an application of Antia–Pearson–Gilchrist nested epidemiology plus Kendall. The public-good model is flagged speculation. Emphasize as synthesis and application. Do not present p_N as an evolutionary theorem.

Emphasize (as application).

1. Macrophage boost is not conserved-potential (R03). Appendix-B / R01 says front-loading birth at fixed total potential does not raise expected size. The plague strategy is *not* that comparison: the macrophage is a different environment. R01 does not kill RQ1.
2. Nested BDC → slightly supercritical BD → binary host-to-host GW (R06, R08). Architecture is Antia et al. 2003 / Pearson / Gilchrist. Algebra is Kendall $(\nu/\beta)^N$ mixed over the burst size. The displayed $\mathbb{E}[Z]$ likely has a parenthesis error (should be $2(1 - p_0)$).
3. Public-good accumulation (R13, R15). Neighbours: Asker et al. 2023; siderophores / yersini-abactin. Known resistance is PhoP/Q, RovA, YCV. Inoculum K (R14) is not computed. Keep as a possible mechanism, not a result.

Do not claim. Mean invariance under conserved replicative density (R01) — Kendall; Pearson conserved-yield. Two-phase pathogenesis (R04, R11) — Cavanaugh 1959; Ke 2013. Neutrophil-fraction threshold p_N (R12): elementary rearrangement; Ke already answers the switch with temperature; reduced form as written has a **sign error** under $\mu_N > \mu_E > \mu_M$. Intended threshold is $(\mu_E - \mu_M)/(\mu_N - \mu_M)$. Two-phase PGF (R07) — branching property.

ID	Short name	Class	Nearest literature
R01	Conserved-replicative-density mean invariance	ALREADY	Kendall 1948; Pearson 2011 conserved yield
R02	Exceptions (non-conservation; Matthew effect)	NOT A RESULT	Named, not developed
R03	Macrophage boost $\neq$ conserved-potential	EXTENDS (remark)	Cavanaugh; Pujol–Bliska; Ke 2013
R04	Two-phase pathogenesis argument	ALREADY	Same
R05	Phage-adaptation decaying extracellular death	EXTENDS	Speculative mechanism
R06	Host-to-host GW embedding	EXTENDS (application)	Whittle/Bartlett; Antia 2003; Pearson; Gilchrist–Sasaki
R07	Two-phase PGF	ALREADY	Branching property
R08	BDC-to-BD extinction and $\mathbb{E}[Z]$	EXTENDS (weak)	Kendall $(\nu/\beta)^N$; check parenthesis
R09	Four within-host variants	EXTENDS	Comparison, not a theorem
R10	Numerical: δ controls burst size and $\mathbb{E}[Z]$	ALREADY (as a lesson)	Haldane/Kendall near-critical founder plot
R11	Quadrant rate ordering	ALREADY	Standard $\mu_N > \mu_E > \mu_M$
R12	Neutrophil-fraction threshold p_N	ALREADY (caricature of Ke)	Ke 2013 temperature cue; sign error
R13	Public-good accumulation model	EXTENDS (speculation)	Asker 2023; siderophores; local public goods
R14	Critical inoculum K	NOT A RESULT	Not computed
R15	Confined public good as group advantage	EXTENDS (speculation)	Nadell/Drescher/Foster

Suggested wording.

The mean-invariance result of the branching appendix is Kendall’s: at conserved replicative potential, front-loading birth does not raise expected size. That comparison is the wrong one for *Yersinia pestis*. The macrophage is a different environment, not a rearrangement of the same potential (Cavanaugh 1959; Ke, Chen and Yang 2013). Embedding a BDC intracellular burst in a slightly supercritical extracellular birth–death process, and that in a host-to-host Galton–Watson process, is the standard nested-epidemic construction (Antia et al. 2003; Pearson et al. 2011) applied to this pathogen. The neutrophil-fraction threshold is a caricature of a switch the experimental literature attributes to temperature and vacuolar cues, not to leukocyte proportions.

Why it matters biologically. RQ1 of the thesis. The strategy is real and famous. The mathematical work that bears on it is: (i) near-critical founder sensitivity (Chapters 1 and 10, classical); (ii) BDC burst size as the macrophage dump (Chapters 4–5, mostly KT plus W_t); (iii) whether bursting then helps against depletable neutrophils/complement (Chapter 6 L , Chapter 8

reversal). The conclusion should point at those, not at p_N .